DSNet: Joint Semantic Learning for Object Detection in Inclement Weather Conditions

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Abstract—In the past half of the decade, object detection approaches based on the convolutional neural network have been widely studied and successfully applied in many computer vision applications. However, detecting objects in inclement weather conditions remains a major challenge because of poor visibility. In this article, we address the object detection problem in the presence of fog by introducing a novel dual-subnet network (DSNet) that can be trained end-to-end and jointly learn three tasks: visibility enhancement, object classification, and object localization. DSNet attains complete performance improvement by including two subnetworks: detection subnet and restoration subnet. We employ RetinaNet as a backbone network (also called detection subnet), which is responsible for learning to classify and locate objects. The restoration subnet is designed by sharing feature extraction layers with the detection subnet and adopting a feature recovery (FR) module for visibility enhancement. Experimental results show that our DSNet achieved 50.84 percent mean average precision (mAP) on a synthetic foggy dataset that we composed and 41.91 percent mAP on a public natural foggy dataset (Foggy Driving dataset), outperforming many state-of-the-art object detectors and combination models between dehazing and detection methods while maintaining a high speed.

Index Terms—Object detection, dual-subnet network, joint learning, multi-task learning, CNN

1 Introduction

In RECENT years, the development of intelligent autonomous vehicles (IAVs) has attracted considerable research attention since it has the potential to assist humans in their driving, reduce the number of traffic accidents, and change cityscapes. The design of a robust and dependable IAV necessitates the integration of many important functions, including but not limited to object detection [1], [2], tracking [3], [4], and human-machine interaction [5], [6].

Object detection plays an essential role in IAVs because it not only determines the category, to which each object belongs and locates objects in a given image, but also assists the system with safe navigation in complex driving environments. Numerous approaches and techniques for object detection have been introduced and have achieved impressive performances in recent years, especially methods using deep convolutional neural networks (CNNs) [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18].

Although these detectors show high performance with normal images, they usually fail to detect objects in inclement weather conditions, especially in fog, one of the most common weather phenomena encountered in driving scenarios. The reason for this is that the specific light spectra between the objects being captured and the camera are absorbed and scattered by suspended tiny water droplets, ice crystals, dust, and other particles, which degrade the features extracted from these images for the purpose of object detection.

Traditionally, to improve object detection performance in the presence of haze, enhancing the visibility of hazy images has been used as a preprocessing step. Image dehazing can benefit both human visual perception for image quality and many systems that must operate under different weather conditions, such as traffic surveillance systems, intelligent vehicles, and outdoor object recognition systems. However, using images preprocessed by a better dehazing model in terms of restoration performance as an input of the object detector does not always guarantee improved object detection performance [19]. Furthermore, combining the models used by dehazing and detection methods can slow the detection speed because of the additional dehazing task.

In this paper, we propose a novel object detection approach called dual-subnet network (DSNet), which is based on multi-task learning to cope with the problem of detecting objects in poor visibility conditions. To attain this objective, DSNet employs one of the best object detectors, namely, RetinaNet [15] as a backbone network (denoted as detection subnetwork) and proposes a feature recovery (FR) module to this base architecture to construct a restoration subnetwork for visibility enhancement, as shown in Fig. 1. The detection and restoration subnetworks are further presented in Section 3. We train DSNet in an end-to-end fashion to simultaneously learn about visibility enhancement, object classification, and object localization. Following the joint optimization scheme, clean features generated by the restoration subnetwork from hazy input images for visibility enhancement can be shared to learn better object classification and object localization in the detection subnetwork, thus helping DSNet

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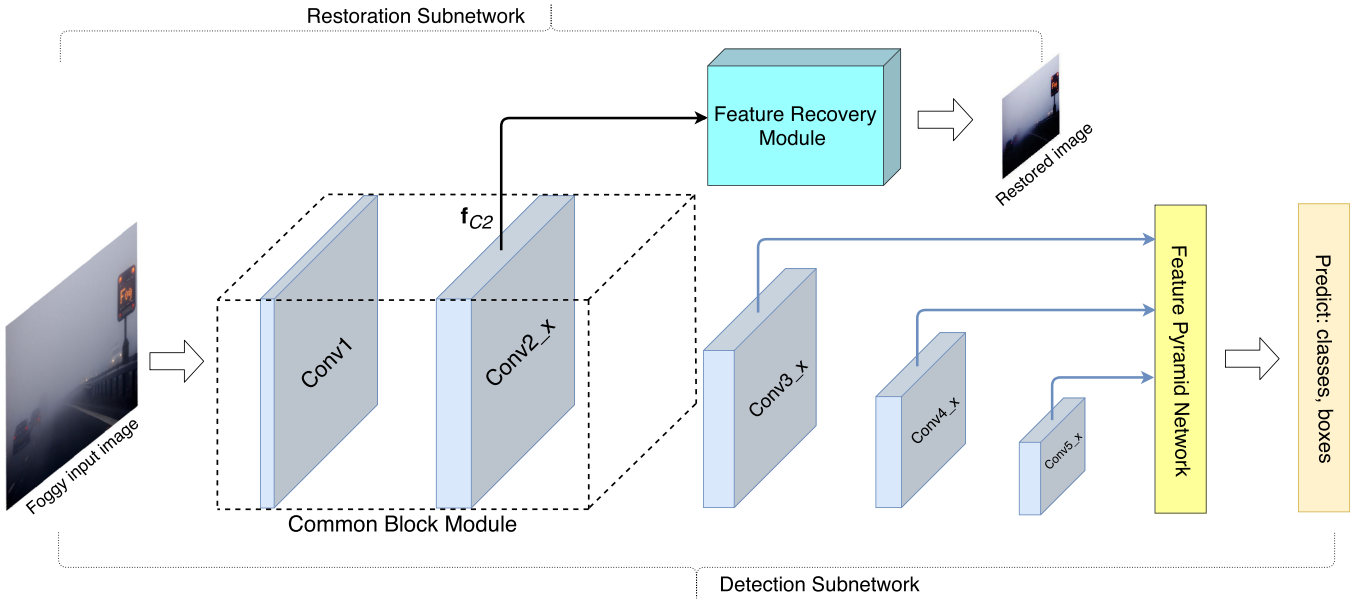


Fig. 1. Architecture of the proposed DSNet. The framework consists of two subnetworks: the Restoration subnetwork and the Detection subnetwork. The Feature Recovery module is only activated during the training phase, and the Restored image is identical to the Foggy input image in size.

boost the object detection performance in inclement weather conditions.

The main features of our proposed method are summarized as follows:

1) The proposed FR module cooperates learning with the detection subnetwork to fulfill the object definition for improving the accuracy of object detection in the poor visibility conditions.
2) Object detection is accomplished by jointly optimizing visibility enhancement, object classification, and object localization.
3) The high speed of the proposed model is preserved compared with the base detection network.

To prove the object detection efficiency of our approach, we evaluate the proposed model on synthetic and natural foggy datasets. Quantitative and qualitative evaluation results show that our proposed approach surpasses the accuracy of current state-of-the-art object detectors and the cascade of dehazing and detection models.

The rest of this paper is organized as follows: Section 2 provides related works including object detection, haze removal, and multi-task learning. Section 3 describes the proposed object detection method, which is used in foggy images. Section 4 presents the comprehensive experimental results of our method compared with other methods. Section 5 concludes our work.

2 Related Work

Many object detection, image dehazing, and multi-task learning methods have been introduced in the literature. In this section, we briefly review some current methods and focus our attention on deep learning-based approaches that have demonstrated remarkable results for these fields.

2.1 Object Detection

Object detection based on convolution neural networks (CNNs) has gained extensive interest to improve performance and can be divided into two main categories [20]: 1) region proposal-based approaches [7], [8], [9], [10] and 2) regression/classification-based approaches [11], [12], [13], [14], [15], [16], [17].

Under the methods belonging to the first category, region proposals with CNNs (R-CNN family) [7], [8], [9] and region-based fully convolution networks (R-FCN) [10] proposed approaches based on the region proposal method to create regions of interest (RoIs) for object detection. R-CNN [7] takes an input image and generates region proposals using an algorithm that calculates hierarchical grouping of similar regions with respect to many compatible elements such as texture, color, size, shape, and so on, which is known as selective search [21], then employs a CNN to extract features for each proposal, and finally adopts support vector machine (SVM) to perform classification. R-CNN showed dramatic performance improvement in object detection and semantic segmentation compared with previous methods [22], [23]. Although R-CNN has achieved impressive accuracy, it still consumes a considerable amount of time during training and inference.

Unlike R-CNN, instead of extracting feature for each region proposal, fast R-CNN [8] uses a CNN to extract features in the whole image, supplements a RoI pooling layer to produce a fixed-length feature vector from each region proposal, and replaces SVM classifier with a softmax layer for classification. Inspired by R-CNN and Fast R-CNN, Ren *et al.* [9] proposed Faster R-CNN by introducing an additional region proposal network (RPN) to replace the selective search algorithm for improving both accuracy and speed. The RPN is a fully convolutional network that takes the feature maps from the fast R-CNN network as input and outputs a set of candidate bounding box coordinates and their score. Dai *et al.* [10] proposed the R-FCN method, which employs a deep residual learning network with 101 layers (ResNet-101) [24] as a backbone network and adds a convolutional layer called position-sensitive RoI pooling layer to the last convolutional block of this network to generate k^2

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position-sensitive score maps with a grid of $k \times k$ for each class. This process helps R-FCN achieve higher speed compared with Faster R-CNN because the computation amount needed for each RoI is significantly reduced.

Under the methods belonging to the second category, YOLO family [11], [12], [13], single shot multiBox detector (SSD) [14] and RetinaNet [15] proposed approaches based on the regression problem, in which bounding box coordinates and class probabilities are predicted directly from images by a single CNN. To achieve real-time detection, Redmon *et al.* [11] proposed YOLO model, which adopts 24 convolutional layers and 2 fully connected layers that inspired by GoogleNet [25] for end-to-end training and uses the topmost layer to predict bounding boxes, confidence scores for these boxes, and class probabilities. The other members in the YOLO family, namely, YOLO-V2 [12] and YOLO-V3 [13] achieved better detection performance through some improvements, including modification of the backbone architecture for feature extraction, an additional network for multiscale predictions, and changes to the training strategy for joint training. Liu *et al.* [14] introduced a method called SSD, which handles the small object problem by adding several convolutional layers on top of the backbone network VGG16 [26] and using multiscale feature hierarchy within this network to detect objects at different scales. Lin *et al.* [15] observed that during the training of a single object detector, a large disparity exists between foreground and background classes. By introducing focal loss, which is bases on cross-entropy loss for classification training, and adopting feature pyramid network [27] for multiscale predictions, Lin's method, namely, RetinaNet demonstrated superior accuracy.

2.2 Image Dehazing

The formation of a hazy image, as described by the atmospheric scattering model, was first introduced by the authors of [28], and was subsequently improved by Nayar *et al.* [29] and Narasimhan *et al.* [30]. The atmospheric scattering model can be written as:

$$I(x) = J(x)t(x) + \alpha(1 - t(x)), \tag{1}$$

where $I(x)$ is the hazy image that is captured by the camera, $J(x)$ is the scene radiance that can be regarded as a haze-free image, α represents the global atmospheric light, and $t(x)$ represents the medium transmission, which is defined as:

$$t(x) = e^{-\beta d(x)}, \tag{2}$$

where β denotes the scattering coefficient of the atmosphere, and $d(x)$ denotes the scene depth.

Based on the atmospheric scattering model, numerous dehazing approaches have been presented for visibility restoration of degraded images to improve the performance of systems that must perform under inclement weather conditions. Early dehazing methods often compute intermediate haze-relevant features based on many techniques, including, but not limited to, dark channel prior [31], [32], maximum contrast [33], color attenuation [34], and hue disparity [35], to accomplish haze removal.

More recently, given the power and success of deep CNN in computer vision tasks, dehazing approaches with deep CNN have been introduced to enhance haze removal performance. To avoid inaccurately estimating the physical parameter from a single image, DehazeNet [36] restores the visibility of hazy images by presenting an end-to-end CNN to produce a medium transmission map from an entire image, which is then applied to the atmospheric scattering model to retrieve haze-free images. MSCNN [37] proposed using two CNNs that have similar architecture called coarse-scale net and fine-scale net. Both these networks contain four components including convolution, max-pooling, up-sampling and linear combination for restoring the visibility of hazy images. First, MSCNN uses the coarse-scale net to generate the rough scene transmission map based on an entire image. Then this transmission map is gradually refined by the fine-scale net. Finally, the refined transmission is used to produce haze-free image by applying the formula (1). Li *et al.* [38] perceived that computing the medium transmission and atmospheric light separately can result in accumulated errors during image restoration. They proposed an all-in-one dehazing CNN called AOD-Net, where the hazy image is taken as the input and the dehazed image is generated as the output of the model. Zhang *et al.* [39] proposed a single image dehazing model called DCPDN, in which the transmission map, atmospheric light, and dehazed image are simultaneously estimated during training. Transmission estimation is performed by the use of multi-level features, atmospheric light estimation is accomplished by employing an 8-block U-net [40] structure, and a dehazed image is produced using an atmospheric scattering model.

2.3 Multi-Task Learning

Multi-task learning is concerned with multiple related tasks that are learned, simultaneously in which what is learned from one task can be beneficial to the other tasks [41]. Many multi-task learning methods have been proposed and proven to be effective for various deep learning applications in the computer vision field. Kendall *et al.* [42] introduced a method that weighs multiple loss functions by considering the homoscedastic uncertainty of each task to perform joint reasoning on three tasks, including depth regression, semantic, and instance segmentation. Liao *et al.* [43] presented a convolutional neural network for learning semantic segmentation and classification at the same time to enhance scene understanding in robotics applications. In [44], the authors applied multi-task learning successfully in a real-time autonomous driving application, wherein classification, detection, and semantic segmentation tasks are incorporated into a model called MultiNet that uses three subnetworks. The approaches used by Lin *et al.* [15] and Kokkinos [45] have also been explored to simultaneously learn regression and classification tasks within a single architecture for object detection.

In this work, unlike the method reported in [15] and [45], our DSNet model jointly learns three tasks, including visibility enhancement, object classification, and object localization for the detection of objects in poor weather conditions. To the best of our knowledge, we are the first research group to demonstrate that object classification, and localization can benefit from joint learning with visibility enhancement to cope with the problem of object detection in degraded images.

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3 Proposed Method

In this section, we introduce a novel dual-subnet network (DSNet) that can significantly improve the accuracy of object detection in inclement weather conditions while maintaining fast prediction running time. Our approach accomplishes this goal by jointly learning three tasks, namely, visibility enhancement, object classification, and object localization. The proposed DSNet consists of two subnetworks: 1) a detection subnet and 2) a restoration subnet. The detection subnet is employed by utilizing a CNN, namely, RetinaNet [15], which shares a common block (CB) module with the restoration subnet and is responsible for object classification and object localization. The restoration subnet is designed by attaching a proposed feature recovery (FR) module to the CB module for visibility enhancement, as shown in Fig. 1. These two subnetworks share the CB module to ensure that the clean features ($\mathbf{f}_{C2}$) produced at this module can be used in both subnetworks during joint learning. DSNet can be trained end-to-end and objects can be predicted by using the detection subnet.

3.1 Detection Subnetwork

As discussed in Section 2, many CNN-based object detectors have been presented to improve detection performance. However, these methods evaluate various candidate locations in each image despite the fact that only a few locations actually contain objects. Therefore, the existing methods usually encounter a massive class-imbalance problem, which can lead to degenerated model results due to the predominance of easily classified background examples during training.

RetinaNet [15] proposes the use of focal loss to down-weight the loss related to well-classified examples and puts greater focus on learning hard-to-classify examples to realize a balance between classes during training. In addition, RetinaNet employs a feature pyramid network (FPN) [27] that provides a top–down pathway and lateral connections that enable the construction of higher resolution layers from a rich semantic layer that can significantly improve the accuracy of detecting small objects in the presence of haze. Because deeper layers contain rich semantic information but lack location information after pooling, lateral connections between deeper layers and corresponding shallower layers are added to enrich the location information, and improve accuracy. For these reasons, RetinaNet has the potential to cope with the challenge of detecting objects in poor visibility conditions. Thus, we adopt RetinaNet as a backbone network, which shares some feature extraction layers called the CB module with the restoration subnetwork and unites with this subnetwork during learning to accomplish object detection (the CB module is described in detail in Section 3.2). Given the many available RetinaNet models, we choose RetinaNet with ResNet-50 [24] and an input image scale of 500 (denoted as RetinaNet or detection subnetwork) in our model.

To detect objects efficiently, the detection subnetwork follows a particular strategy. First, the ResNet-50 architecture is employed to extract the feature maps of a whole image. Next, at the top of the ResNet-50 architecture, an FPN [27] is built and appended to construct multiscale features to cope with objects at different scales. Finally, object classification and localization tasks are completed by adding small fully convolutional networks (FCNs) to each FPN level to implement convolutional object classification and bounding box regression.

To train classification, the detection subnetwork adopts focal loss [15] with a λ_c - balanced variant, which is described as:

$$L_{cls}(p_c) = -\lambda_c(1-p_c)^{\gamma}\log(p_c), \tag{3}$$

where $\lambda_c \in [0,1]$ for class 1 and $1-\lambda_c$ for class -1, γ is a tunable focusing parameter ($\gamma \geq 0$), and p_c is defined as:

$$p_c = \begin{cases} p & \text{if } y = 1 \\ 1-p & \text{otherwise,} \end{cases} \tag{4}$$

where y specifies the ground-truth class ($y \in \{\pm 1\}$), p is the probability for the class with label $y = 1$ that is estimated by the model ($p \in [0, 1]$).

For localization, the detection subnetwork adopts $Smooth_{L1}$ loss [8] between the predicted box (k) and the ground truth box (g) parameters. The matching pairs between the anchor box (l) and the ground truth box (g) are denoted as $(l^m, g^m)_{m=1,2,\ldots q}$, where q is the number of matches. For each anchor box with a match, a ground truth box regression is defined as $\hat{g}^m = (\hat{g}^m_x, \hat{g}^m_y, \hat{g}^m_w, \hat{g}^m_h)$, and a corresponding predicted box $k^m = (k^m_x, k^m_y, k^m_w, k^m_h)$, in which x, y, w, and h are the two center coordinates of the box, width, and height, respectively. The localization loss is described as:

$$L_{loc}(k^m_i, g^m_i) = \sum_{i \in \{x,y,w,h\}} smooth_{L1}(k^m_i - \hat{g}^m_i), \tag{5}$$

where

$$\hat{g}^m_x = \frac{g^m_x - l^m_x}{l^m_w},\ \hat{g}^m_y = \frac{g^m_y - l^m_y}{l^m_h},\ \hat{g}^m_w = \log\left(\frac{g^m_w}{l^m_w}\right),\ \hat{g}^m_h = \log\left(\frac{g^m_h}{l^m_h}\right).$$

3.2 Restoration Subnetwork

In our DSNet, the restoration subnetwork is responsible for generating the $\mathbf{f}_{C2}$ and shares these features with the detection subnetwork during joint learning to improve the accuracy of object detection in poor visibility conditions. This subnetwork completes the goal on the basis of a transformed atmospheric scattering model and uses two modules: 1) a CB module and 2) an FR module, as shown in Fig. 2.

Transformed formula: The scene radiance can be gained from the atmospheric scattering model in Eq. (1) as:

$$J(x) = \frac{1}{t(x)}I(x) - \alpha\frac{1}{t(x)} + \alpha. \tag{6}$$

As mentioned in Section 2, the dehazing approaches are based on the transformed formula in (6) to enhance visibility in hazy images, in which the medium transmission ($t(x)$) and the global atmospheric light (α) are evaluated separately. However, this approach usually necessitates more steps and increases the complexity of the model.

Inspired by [38], the restoration subnet incorporates $t(x)$ and α into one variable during visibility enhancement. To do this, the formula in (6) is rewritten as:

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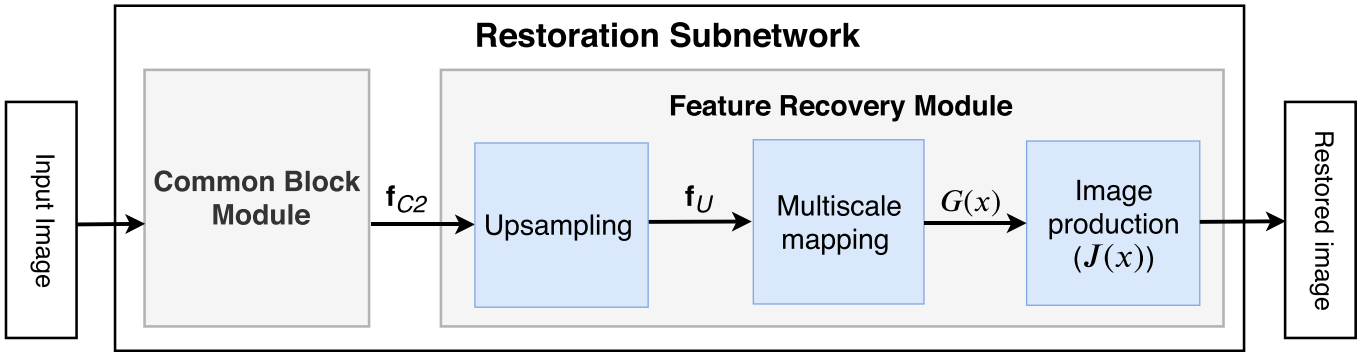


Fig. 2. Flowchart of the Restoration subnetwork. Notably, because the framework only focuses on the object detection task, the Feature Recovery Module is only activated during training to produce restored images for the proposed multi-task learning procedure.

$$J(x) = G(x)I(x) - G(x) + 1, \tag{7}$$

where

$$G(x) = \frac{\frac{1}{t(x)}(I(x) - \alpha) + (\alpha - 1)}{I(x) - 1}. \tag{8}$$

The estimation of $G(x)$ and calculation of $J(x)$ through our model are expressed in more detail in Section 3.2.2.

3.2.1 CB Module

In our model, the CB module extracts the features of the input image that encompass important information for simultaneously learning visibility enhancement, object classification, and object localization.

Instead of being designed separately, the CB module is equipped with some residual blocks of the detection subnet. This approach allows us to preserve a simple architecture of the proposed network. In particular, 16 residual blocks are divided into four residual stages (denoted as C2, C3, C4, and C5) in the detection subnet. Given that features from shallow layers may contain more spatial information that is beneficial to visibility enhancement whereas those of deeper layers are reduced during pooling, we select the first 10 convolutional layers of the detection subnet to compose the CB module and take C2 as the output of this module, as depicted in Fig. 1. The resulting feature maps are simultaneously delivered to the FR module for visibility enhancement and Conv3_x for detecting objects.

3.2.2 FR Module

The features of the input image extracted by the CB module can be degraded by haze, thereby leading to poor object detection performance. To restore $\mathbf{f}_{C2}$, which are shared to effectively reason about object detection and object classification during joint learning, the restoration subnet proposes using an FR module. This FR module is inspired by [38] and consists of three submodules: 1) upsampling submodule, 2) multiscale mapping (MM) submodule, and 3) image production (IP) submodule, as can be seen in Fig. 2.

Upsampling Submodule. In our restoration subnet, the size of the output image and the input image is equal. However, after extracting features by using the CB module, the size of $\mathbf{f}_{C2}$ is one-fourth of the input size. Consequently, the FR module employs the upsampling submodule to match the resolutions of $\mathbf{f}_{C2}$ and the input image.

In [46], the bilinear interpolation technique has been proven important the CNN-based haze removal approach, where the pooled feature maps are bilinearly upsampled to prepare for generating the dehazed output. Inspired by this method, our upsampling submodule first employs a 1×1 convolutional layer to reduce the feature dimensions; in particular, the number of channels of $\mathbf{f}_{C2}$ is shrunk by eight times. Afterwards, a bilinear interpolation is used to enlarge the size of $\mathbf{f}_{C2}$ to the same size as the input images.

MM Submodule. After the resolution of $\mathbf{f}_{C2}$ is increased by utilizing the upsampling submodule, the resulting feature maps called $\mathbf{f}_U$ are delivered to an MM submodule for multiscale feature extraction. Multiscale features have been widely applied to many haze removal methods and are effective for enhancing visibility. For example, in [36], multiscale features for medium transmission estimation are formed by using parallel convolutional layers with different kernel sizes. The authors in [38] proposed an approach that obtains multiscale features by concatenating the features from all network layers to generate the dehazed output in one unified model.

Inspired by the achievements of multiscale feature extraction, the MM submodule employs convolution layers in parallel based on the Inception architecture [25] to extract multiscale features for enhancing visibility. The MM submodule is composed of four parallel convolutions, namely, one 1×1 convolution, one 3×3 convolution, one 5×5 convolution, and one 7×7 convolution with the same number of channels, which is set at 4. The resulting features are fused and followed by another 3×3 convolution to estimate $G(x)$ which contains both the medium transmission $(t(x))$ and the global atmospheric light (α), as can be seen in Eq. (8).

IP Submodule. As the final step of the restoration subnet, the scene radiance is recovered by using the IP submodule. The IP submodule takes $G(x)$ as input and adopts an element-wise multiplication layer, an element-wise subtraction layer, and an element-wise addition layer to calculate the transformed formula in (7).

To train visibility enhancement, the restoration subnetwork employs mean square error (MSE) loss, which is described as:

$$L_{ve} = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{Y}_j - \hat{\mathbf{Y}}_j)^2, \tag{9}$$

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Fig. 3. Example images and annotations from the FOD dataset. The purple and yellow boxes denote the ground-truth bounding boxes of the car and person categories, respectively.

where n is the batch size, $\mathbf{Y}_j$ is the ground truth image, and $\hat{\mathbf{Y}}_j$ is the estimated recovery image.

We emphasize that, although the restoration subnetwork can directly produce haze-free images, its purpose is not to generate the haze-free image as input of the detection subnetwork, but to generate clean features ($\mathbf{f}_{C2}$) from the CB module through learning the visibility enhancement task. As a result, the improved detection performance of the proposed model is achieved by jointly optimizing visibility enhancement, object classification, and object localization.

4 Experimental Results

This section summarizes the experimental results of our method for object detection in inclement weather conditions. To conduct experiments, a dataset for detecting objects in fog is composed, called the fog object detection (FOD) dataset. Two scenarios of comprehensive comparison between our method and object detection methods are implemented by evaluating the prediction quality and quantity on both synthetic foggy dataset (FOD dataset) and natural foggy dataset (Foggy Driving dataset [47]). The first scenario compares our proposed DSNet with four state-of-the-art object detection models, including RetinaNet [15], YOLO-V3 [13], SSD512 [14], and Faster R-CNN [9]. In the second scenario, our DSNet is compared with the combination of RetinaNet and three state-of-the-art dehazing models: 1) AOD-Net [38] with RetinaNet (denoted as AOD-RetinaNet), 2) DCPDN [39] with RetinaNet (denoted as DCPDN-RetinaNet), and 3) CAP [34] with RetinaNet (denoted as CAP-RetinaNet).

We use Keras for all our experiments and run it on a system with an Intel Core i5-8500 CPU @3.00 GHz, 16 GB RAM, and NVIDIA GeForce GTX 1070 Ti card.

4.1 Datasets

4.1.1 FOD Dataset

Few datasets have been published for object detection in inclement weather conditions. Boyi *et al*. [19] introduced the RTTS dataset, which involves 4,322 natural hazy images with five annotated object categories, namely, bus, bicycle, car, motorbike, and person. This dataset is used for testing purposes only. To evaluate object detection algorithms in real-world rainy weather conditions, Li [48] introduced two sets including rainy in driving set and rain in surveillance set. Each set comprises about 2,000 images and five objects (person, bus, bicycle, car, motorcycle) are labeled bounding boxes. In [47], the authors collected a Foggy Cityscapes-refined dataset, which consists of 550 images (498 images of the training set, 52 images of the validation set). However, these datasets are too small in scale for deep CNN training strategies to improve object detection performance.

To conduct all our experiments, we compose an FOD dataset by collecting images from the public Foggy Cityscapes dataset [47] and labeling two classes, namely, person and car. In the FOD dataset, 6,000 synthetic foggy images are carefully selected from three versions of Foggy Cityscapes: heavy fog, medium fog, and light fog, which have attenuation coefficient values of 0.02, 0.01 and 0.005, respectively. A total of 119,979 object instances are labeled following the Cityscapes [49] annotation protocol, where the number of person instances is 60,279 and those of car instances is 59,700. We split the FOD dataset into three sets: training set, validation set, and test set; no repetition of given images among each set. Example images with bounding box annotations overlaid and the statistics of the FOD dataset are displayed in Fig. 3 and Table 1, respectively. In addition, given that the proposed FOD dataset is based on the Cityscapes

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TABLE 1
Statistic of the FOD Dataset

Set	Number of Images	Number of person instances	Number of car instances
Training	4,200	41,873	41,699
Validation	600	6,167	6191
Test	1,200	12,239	11,810
Total	6,000	60,279	59,700

dataset [49], we can use the ground-truth images to train the restoration subnet.

4.1.2 Foggy Driving Dataset

The Foggy Driving dataset [47] is a natural foggy dataset that is used for testing object detection and semantic scene understanding. It consists of more than 500 vehicle instances (car, truck, bus, train, motorcycle, and bicycle) and about 300 human instances (person and rider) that are annotated from 101 natural foggy images.

Although the Foggy Driving dataset has a small number of images, it is a challenging dataset because of highly complex scenes in the fog.

4.2 Implementation Details

Training Strategy. We apply a fine-tuning approach to our DSNet for jointly training the detection subnetwork (RetinaNet) and the restoration subnetwork. In the first step, we train ResNet-50 [24] on ImageNet [50] because the proposed model is based on the ResNet architecture as a backbone for computing feature map extraction. Then, the detection subnetwork is trained on COCO [51] for objects prediction. Instead of training from scratch, this step is completed by using the well-trained model in [15]. In the second step, our model is trained end to end on the FOD dataset for simultaneously learning visibility enhancement, object classification, and object localization.

Loss Function. The L_{cls} loss defined in Eq. (3), L_{loc} loss defined in Eq. (5) and L_{ve} loss defined in Eq. (9) are summed as the loss for the joint training.

Initialization. The weights of the restoration subnetwork are initialized using Gaussian random variables. With the detection subnetwork, instead of randomly initializing weights, the trained RetinaNet [15] model on the COCO dataset is adopted and the fully trained weights of this model are subsampled from 80 classes to two classes (person and car) for fine-tuning.

Optimization. The proposed model is trained using the Adam optimizer [52]. We set 100 epochs, a weight decay of 0.0001, and a batch size of 4 in training. A learning rate of 10^{-4} is applied for the first 60 epochs, a learning rate of 10^{-5} is used for the next 20 epochs, and 10^{-6} is used for the final epochs. The early stopping training approach is used to avoid overfitting.

Evaluation. The performance of object detection in foggy images is evaluated using the PASCAL VOC metric [53], which is a well-known standard measure that is expressed as follows.

For each object category, the intersection over union between a predicted bounding box and a ground-truth bounding box is defined as:

$$IoU = \frac{(\Omega_g \cap \Omega_p)}{(\Omega_g \cup \Omega_p)}, \tag{10}$$

where Ω_g is the area of ground-truth bounding and Ω_p is the area of predicted bounding box.

A detection is considered to be correct (true positive) if the IoU score (threshold) is greater than 0.5, otherwise it is considered wrong (false positive).

To measure the performance of the object detectors, an AP metric is adopted, which is based on the precision-–recall curve (PRC). The PRC indicates the relationship between precision (denoted as P_r) and recall (denoted as R_e) for different thresholds, in which P_r and R_e are expressed as follows:

$$P_r = \frac{TP}{TP + FP}, \tag{11}$$

$$R_e = \frac{TP}{TP + FN}, \tag{12}$$

where TP, FP and FN are the number of true positives, false positives, and false negatives, respectively.

The AP that presents the area under the PRC is defined as follows:

$$AP = \int_0^1 P_r(R_e) d_{R_e}. \tag{13}$$

An object detector that has a higher AP value means that it has better performance and vice versa.

The mean average precision (mAP), which is the average AP value for all object categories is calculated as follows:

$$mAP = \frac{1}{M} \sum_{z=1}^{M} AP_z, \tag{14}$$

where M is the number of object categories.

4.3 Results on the FOD Dataset and the Foggy Driving Dataset

4.3.1 Ablation Study on FOD Dataset

In this subsection, to determine the best design for the proposed model, we describe our model runs using many different settings on the FOD dataset, including the structure of the CB module, upsampling technique, and the separate use of the detection, and restoration subnetworks.

Structure of CB Module and Upsampling Technique. We investigate four different structures to effectively select the factors of the CB module. As depicted in Fig. 1, the first structure is composed of the first 10 feature extraction layers of the detection subnet (RetinaNet [15]). We call this version CB_C2 because its output corresponds to the output of residual stage C2. The second structure is called CB_C3, which utilizes four more residual blocks (Conv3_x) than the CB_C2 module to learn deeper features and takes C3 as the output of the module. We continue exploring the deeper architecture for the CB module with the third and the final implementations called CB_C4 and CB_C5, respectively. CB_C4 discards the last three residual blocks (Conv5_x) of ResNet to make the output at C4, and CB_C5 uses all 16 blocks to output feature maps at C5.

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TABLE 2
Effects of the CB Module and Upsampling Submodule in the DSNet Model on the FOD Test Set

CB module	Upsampling submodule	
	Bilinear interpolation (mAP)	Deconvolution (mAP)
CB_C2	50.84	50.51
CB_C3	50.21	49.77
CB_C4	49.47	48.86
CB_C5	47.76	47.30

Upsampling technique: We explored two techniques, i.e., bilinear interpolation and deconvolution, to increase the size of the feature maps to match the resolutions of the input and output of the restoration subnetwork.

Table 2 presents a comparison of different structures of the CB module and the upsampling techniques used in our DSNet model. The second column of Table 2 shows the results obtained when the bilinear interpolation technique is applied. The mAP achieved when using the CB_C2 module is 50.84 percent, which is 0.63, 1.37, and 3.08 percent higher than those obtained when using the CB_C3, CB_C4, and CB_C5 modules, respectively. The third column of Table 2 shows the results obtained when the deconvolution technique is employed. The mAP achieved when using the CB_C2 module is 50.51 percent, which is 0.74, 1.65, and 3.21 percent higher than those obtained when using the CB_C3, CB_C4, and CB_C5 modules, respectively.

These results can be explained as follows: When using the CB module with the deeper architecture, the amount of rich spatial information that can be useful for visibility enhancement is reduced. Consequently, the restoration subnet may not work as well to improve the visibility of foggy images, thereby leading to the poor quality of $\mathbf{f}_{C2}$ generated at the CB module for learning object classification and localization.

Because the bilinear interpolation technique and CB_C2 module yielded the best model results, they were selected for designing our DSNet model.

Impact of Restoration Subnetwork. In this subsection, we conduct an ablation study to investigate the impact of the Restoration subnetwork on the proposed model. The experimental results are shown in Fig. 4. When the Restoration subnetwork is removed, the model includes only the Detection subnetwork (denoted as DSNet-1) with localization loss defined in Eq. (5) and Focal loss defined in Eq. (3). In this case, the network architecture is identical to that of RetinaNet [15], and obtains an average precision (AP) for cars and persons of 55 and 40.96 percent, respectively.

Fig. 4 reveals that the accuracy rates obtained by using the DSNet model are 58.01 percent AP for the cars, and 43.67 percent AP for the persons, which are 3.01 and 2.71 percent higher than those provided by the DSNet-1 model. The main reason behind such performance improvement without additional computation is that: during the training process, in order to let FR module produces a clean haze-free image from $\mathbf{f}_{C2}$, the CB module will learn to extract features that contain more haze-free spatial-related features to minimize the restoration loss. While the feature $\mathbf{f}_{C2}$ contains more haze-free spatial features, the Detection subnetwork is able to use these features to accurately detect objects in hazy images.

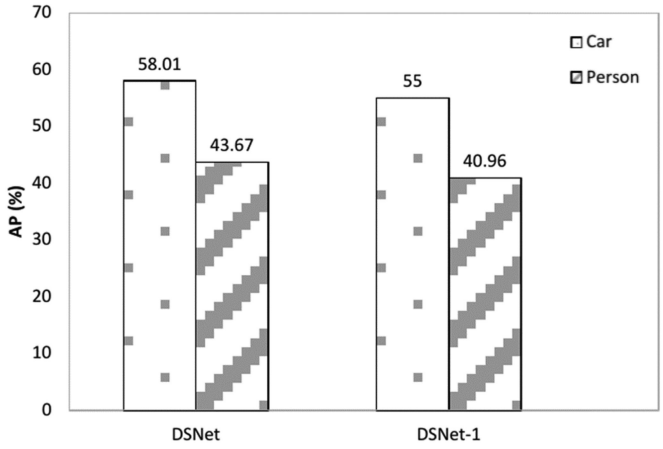


Fig. 4. Effects of the Restoration subnetwork in the proposed model on the FOD test set. DSNet is our model which contains both the Detection and Restoration subnetworks. DSNet-1 denotes the proposed model without using the Restoration subnetwork.

4.3.2 Qualitative and Quantitative Results

Object Detectors. In this subsection, we qualitatively and quantitatively evaluate the performance of our DSNet model, and compare our findings with the performances of the base detector (RetinaNet [15]) and current state-of-the-art object detectors, namely, YOLO-V3 [13], SSD-512 [14], and Faster R-CNN [9] with respect to the FOD dataset and the Foggy Driving dataset.

In our experiment, all of the compared methods were trained by applying the fine-tuning approach. The loss functions and settings of each compared model (i.e., the scales, the aspect of the anchor or default boxes, the backbone network, the classifier, and so on) remained unchanged from the original work, except for Faster R-CNN, which uses Resnet-50 [24] as its base network instead of VGG16 [26]. In addition, although the test is conducted on two different datasets, all of the methods were trained by using only the FOD dataset.

To train RetinaNet, YOLO-V3, and SSD512, we adopted the conventional models reported in [15], [13], and [14], respectively, which were pre-trained on the COCO dataset [51]. Then, we subsampled the classifier weights of these models from 80 classes to 2 classes (person and car), and fine-tuned them on the FOD training set. To train Faster R-CNN, the ResNet-50 backbone was first trained on the ImageNet dataset. Instead of training ResNet-50 from scratch, we utilized the pre-trained model in [50] to fine-tune the Faster R-CNN model on the FOD training set.

The qualitative results for our DSNet, RetinaNet, YOLO-V3, SSD512, and Faster R-CNN in detecting objects in the presence of fog are shown in Fig. 5. DSNet surpasses the other models in accurately detecting persons and cars in the fog.

The detection accuracy of the proposed DSNet and the above mentioned models is quantitatively compared by using the mAP metric. The mAP results of the compared methods are listed in Table 3. The accuracy rate of our method is higher than that of the other methods on both the FOD and Foggy Driving datasets.

The second column of Table 3 presents the comparison results on the FOD test set. DSNet achieves a detection accuracy up to 2.86, 4.08, 5.19, and 9.37 percent higher than

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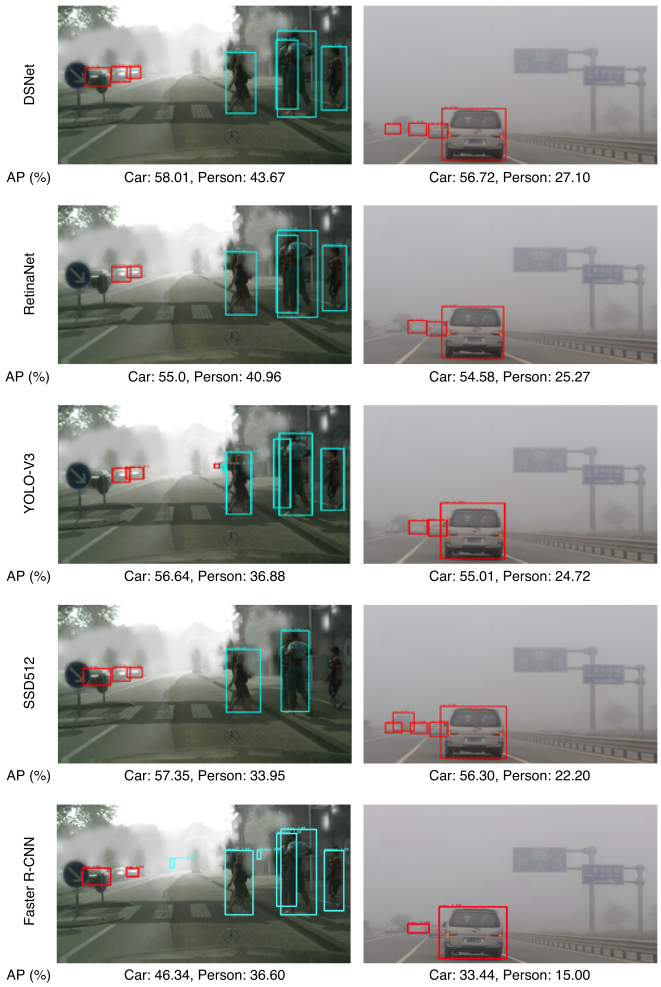


Fig. 5. Detection results of our DSNet and different object detectors on synthetic foggy images (left column), natural foggy image (right column). The red and cyan boxes demonstrate the predicted bounding boxes of the car and person, respectively. Bounding boxes with a score greater than or equal to 0.5 will be drawn.



Fig. 6. Detection results of our DSNet and different combination models on synthetic foggy images (left column), natural foggy image (right column). The red and cyan boxes demonstrate the predicted bounding boxes of the car and person, respectively. Bounding boxes with a score greater than or equal to 0.5 will be drawn.

that achieved by RetinaNet, YOLO-V3, SSD512, and Faster R-CNN respectively.

The third column of Table 3 presents the comparison results on the Foggy Driving dataset. DSNet is 1.99, 2.04, 2.66, and 17.69 percent more accurate than RetinaNet, YOLO-V3, SSD512, and Faster R-CNN, respectively.

Defogging for Detection. To further validate our DSNet model, we apply the dehazing method to the object detection problem in the presence of fog to improve the performance

TABLE 3
The mAP of the Proposed DSNet and Four State-of-the-Art Object Detection Methods on the FOD Test Set and the Foggy Driving Dataset

Methods	FOD Test Set (mAP)	Foggy Driving (mAP)
DSNet	50.84	41.91
RetinaNet [15]	47.98	39.92
YOLO-V3 [13]	46.76	39.87
SSD512 [14]	45.65	39.25
Faster R-CNN [9]	41.47	24.22

Note that all methods were fine-tuned on the FOD dataset, and Faster-RCNN [9] used ResNet-50 [24] as a backbone network instead of VGG16 [26].

and compare it with our proposed method. We select three popular dehazing methods, namely, AOD-Net [38], DCPDN [39], and CAP [34] to combine with the base detection network (RetinaNet [15]) for forming three combination models called AOD-RetinaNet, DCPDN-RetinaNet, and CAP-RetinaNet, respectively.

The performance of the three combination models is obtained by first training AOD-Net, DCPDN, and CAP on the FOD training set to perform fog removal. Then, three fog-free image sets are generated by using the trained AOD-Net, DCPDN, and CAP models on the FOD test set. Afterwards, we train RetinaNet on the ground-truth images from the FOD dataset to perform object detection in clean scenes. Finally, the results of three concatenation models are achieved by applying the trained RetinaNet model to three fog-free image sets.

Fig. 6 shows a qualitative comparison between our DSNet and three combination models in object detection in the presence of fog. We can observe that DSNet more accurately detects objects of interest (person and car) than the other models.

Table 4 shows a quantitative comparison between DSNet, RetinaNet, and three concatenation models on the both FOD and Foggy Driving datasets. The second column of Table 4 lists the mAP results of the compared methods on the FOD test set. Our DSNet is better than RetinaNet, AOD-RetinaNet, CAP-RetinaNet, and DCPDN-RetinaNet by 6.54, 6.11, 6.24, and 9.17 percent, respectively. The third column of Table 4 lists the mAP results of the compared methods on the Foggy

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TABLE 4
The mAP and Speed of the Proposed DSNet, RetinaNet, and Three Concatenation Models on the FOD Test Set and Foggy Driving Dataset

Methods	FOD Test Set (mAP)	Foggy Driving (mAP)	Speed (ms)
DSNet	50.84	41.91	47
RetinaNet [15]	44.30	37.73	47
AOD-RetinaNet	44.73	34.69	151
CAP-RetinaNet	44.60	35.71	155
DCPDN-RetinaNet	41.67	31.32	169

Note that RetinaNet was trained with ground-truth images in the FOD dataset, and the defogging methods (AOD, CAP, DCPDN) were trained with foggy/ground-truth images in the FOD dataset.

Driving dataset. Our DSNet surpasses RetinaNet, AOD-RetinaNet, CAP-RetinaNet, and DCPDN-RetinaNet by 4.18, 7.22, 6.20, and 10.59 percent, respectively.

4.4 Inference Time

In our DSNet model, the detection subnetwork is responsible for joint inference of object classification and object localization in foggy images after simultaneously learning with the restoration subnetwork. Therefore, although DSNet adds additional convolutional layers on the base network, it preserves the high detection speed. The inference speed of the compared methods is listed in the fourth column of Table 4. Our method can run 47 ms per image on a single GTX 1070 Ti GPU and is equal to RetinaNet; it is 104, 108, and 122 ms faster than AOD-RetinaNet, CAP-RetinaNet, and DCPDN-RetinaNet, respectively, while achieving the best accuracy.

5 Conclusion

In this paper, we presented a novel approach to improve object detection performance in inclement weather conditions. Our DSNet model, which can be trained end to end for joint learning of visibility enhancement, object classification and object localization, is composed of two subnetworks, namely, the detection and restoration subnets. The detection subnet is introduced using RetinaNet, and the restoration subnet is designed by attaching the FR module to the last feature extraction layer of the third residual block of the detection network. The experimental results on both synthetic and natural foggy datasets indicate that our proposed approach obtained the most satisfactory detection performance. Qualitative and quantitative evaluations of the compared methods prove that our DSNet is significantly more accurate than the other models while maintaining a high speed.

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